A11.2 Temperature and Rate of Reaction

Kinetic energy • collision frequency • successful collisions • Maxwell–Boltzmann distribution

T-LEVELS

Learning outcomes

- Explain why increasing temperature increases particle kinetic energy and particle speed.
- Explain how higher temperature changes collision frequency and the proportion of successful collisions.
- Explain the corresponding effect of lower temperature on rate of reaction.
- Interpret the effect of temperature using a Maxwell–Boltzmann distribution.
- Apply the complete particle-level explanation to unfamiliar reaction scenarios.

Starter: connect to collision theory

- What two conditions are required for a successful collision?
- What does activation energy represent?
- If particles move faster, what might happen to collision frequency?
- If particles have more kinetic energy, what might happen to the proportion with enough energy to react?

Starter answers

- Sufficient collision energy and correct spatial orientation.
- Activation energy is the minimum collision energy required for a successful reaction.
- Faster-moving particles collide more frequently.
- A greater proportion of particles can have energy equal to or greater than the activation energy.

A11.2.1 Temperature changes kinetic energy

- Temperature is linked to the kinetic energy of particles.
- When temperature increases, particles have greater average kinetic energy.
- Greater kinetic energy means particles move faster.
- Faster movement changes both how often particles collide and the energies of those collisions.

First effect: collision frequency

- At higher temperature, particles move faster through the reaction mixture.
- They encounter other reactant particles more frequently.
- Therefore the frequency of collisions increases.
- More collisions per unit time creates more opportunities for successful collisions.

Second effect: proportion of successful collisions

- At higher temperature, the particle-energy distribution changes.
- A greater proportion of particles have energy equal to or greater than the activation energy.
- Therefore a greater proportion of collisions can satisfy the energy requirement.
- Correct orientation is still required: increased temperature does not remove the orientation condition.

Complete explanation: increasing temperature

- Higher temperature → greater average kinetic energy.
- Particles move faster → collisions occur more frequently.
- A greater proportion of particles have energy ≥ activation energy.
- Therefore successful collisions occur more frequently.
- The rate of reaction increases.

Activity: Temperature and Collision Model

Teacher action

- Model the same particles at lower and higher temperature.
- Separate the explanation into collision frequency and collision success.

Student action

- Use particle diagrams to compare particle speed and collision frequency.
- Explain why the proportion of successful collisions changes.
- Construct the full causal chain from temperature to rate.

Temperature and Collision Model: feedback

- Higher T → greater average kinetic energy → faster particles.
- Faster particles → increased collision frequency.
- Energy distribution changes → greater proportion with E ≥ Ea.
- More successful collisions per unit time → increased reaction rate.

A11.2.2 What happens when temperature decreases?

- Lower temperature means lower average kinetic energy.
- Particles move more slowly.
- Collision frequency decreases.
- A smaller proportion of particles have energy equal to or greater than activation energy.
- Successful collisions occur less frequently, so reaction rate decreases.

Lower temperature: avoid a common shortcut

- It is incomplete to say only 'particles collide less'.
- A full explanation includes both fewer collisions per unit time AND a smaller proportion of collisions with sufficient energy.
- The activation energy of the same reaction has not changed simply because temperature changed.
- Orientation remains a requirement at all temperatures.

Activity: Low Temperature Prediction Challenge

Teacher action

- Model a decrease in temperature using particle motion and energy language.
- Prompt students to include both collision frequency and collision success.

Student action

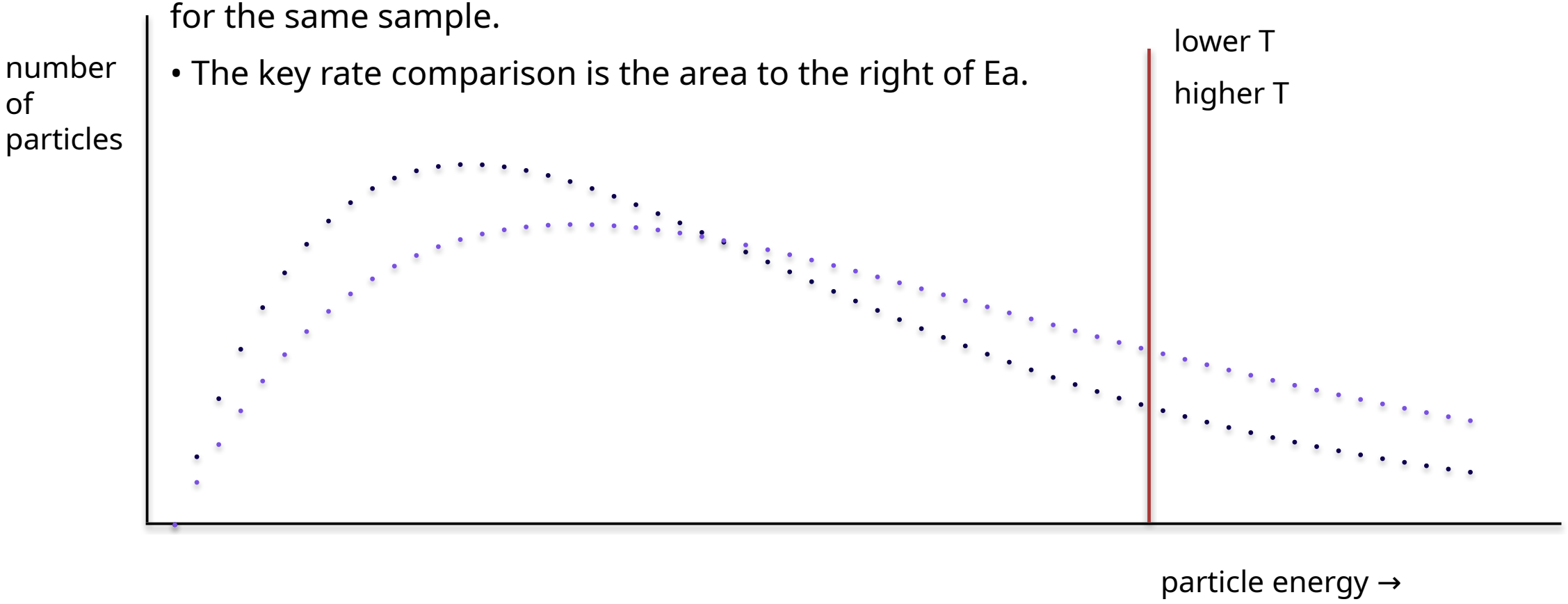
- Predict changes to kinetic energy, particle speed, collision frequency and successful-collision proportion.
- Use these changes to explain the lower rate of reaction.

A11.2.3 Maxwell–Boltzmann distribution: what the graph shows

- The Maxwell–Boltzmann distribution shows how particle energies are distributed in a sample at a particular temperature.
- The horizontal axis represents particle energy.
- The vertical axis represents the number (or fraction) of particles with a given energy.
- The activation energy can be shown as a vertical threshold, Ea.

Maxwell–Boltzmann distributions at two temperatures

- The higher-temperature curve is broader, has a lower peak and is shifted toward higher energies.
- The total number of particles is unchanged, so the total area under each distribution is the same for the same sample.
- The key rate comparison is the area to the right of Ea.



Interpreting the area beyond Ea

- Particles to the right of the activation-energy line have energy ≥ Ea.
- At higher temperature, the area to the right of Ea is larger.
- Therefore a greater proportion of particles have sufficient energy for a successful collision.
- This helps explain the increased reaction rate at higher temperature.

What changes and what does not

- Changes with temperature: shape of the energy distribution, average kinetic energy and proportion with $E \geq Ea$.
- Does not change for the same uncatalysed reaction: the activation-energy threshold itself.
- The graph supports the energy part of collision theory; successful collisions still also require correct orientation.

Activity: Maxwell–Boltzmann Graph Analysis

Teacher action

- Model how to locate Ea and compare the areas to its right.
- Emphasise curve shape: higher T gives lower peak and broader distribution.

Student action

- Compare two distributions at different temperatures.
- Identify which represents the higher temperature and justify.
- Explain how the area beyond Ea changes and link this to reaction rate.

Guided graph interpretation

- Curve A is taller and narrower; Curve B is lower and broader.
- The activation-energy line is fixed.
- Which curve represents the higher temperature?
- Which has the larger proportion of particles with $E \geq Ea$?
- Use the graph to explain which reaction mixture has the greater rate.

Guided graph interpretation: answers

- Curve B represents the higher temperature because the distribution is broader and shifted toward higher energies, with a lower peak.
- Curve B has the larger area to the right of Ea.
- A greater proportion of particles therefore have sufficient energy for successful collisions.
- Together with increased collision frequency, this gives a greater reaction rate.

Independent: Distribution Interpretation Challenge

Teacher action

- Provide unfamiliar Maxwell–Boltzmann distributions with a marked Ea.
- Require graph evidence plus collision-theory reasoning.

Student action

- Identify higher and lower temperature curves.
- Compare the proportion of particles above Ea.
- Explain the predicted rate difference using kinetic energy, collision frequency and successful collisions.

Error Detective

- Student A: 'Heating lowers the activation energy.'
- Student B: 'The peak gets higher when temperature increases.'
- Student C: 'Only collision frequency changes with temperature.'
- Student D: 'Every collision above Ea is automatically successful.'
- Correct each statement using the specification.

Error Detective: answers

- A: Temperature changes the particle-energy distribution; it does not by itself lower Ea for the same reaction.
- B: At higher temperature the distribution becomes broader and the peak becomes lower.
- C: Collision frequency increases and the proportion of particles with $E \geq Ea$ also increases.
- D: Sufficient energy is necessary, but correct spatial orientation is also required.

Exam style application

- A reaction mixture is heated from 298 K to 318 K.
- (a) Explain how the increase in temperature affects particle kinetic energy and collision frequency. [2]
- (b) Explain why the proportion of successful collisions increases. [2]
- (c) Describe how the Maxwell–Boltzmann distribution changes. [3]
- (d) Explain why the reaction rate increases, using collision theory. [3]

Exam style application: answers

- (a) Average kinetic energy increases; particles move faster and collide more frequently.
- (b) A greater proportion of particles have energy equal to or greater than Ea, so more collisions satisfy the energy requirement.
- (c) The distribution becomes broader, its peak becomes lower, and a greater area lies at higher energies/to the right of Ea.
- (d) Collision frequency increases and a greater proportion of collisions have sufficient energy; therefore successful collisions occur more frequently and rate increases. Correct orientation is still required.

Final knowledge check

- Explain the effect of increasing temperature on average kinetic energy.
- Explain the effect on collision frequency.
- Explain the effect on the proportion of particles with $E \geq Ea$.
- Describe the higher-temperature Maxwell–Boltzmann curve.
- Explain the effect of lowering temperature on reaction rate.
- State whether temperature alone changes Ea for the same reaction.